FinRisk AI

AI-Powered Financial Risk Intelligence & Event-Driven Stress Testing

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Problem & Approach

The problem

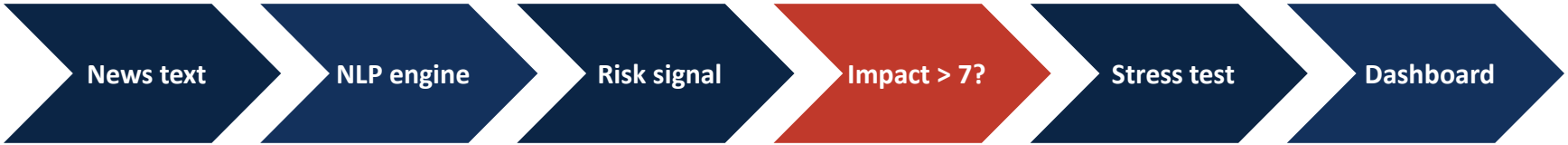
Risk teams drown in unstructured news but need structured signals and a fast answer: what happens to our portfolio?

Our approach

A rule-based NLP engine turns headlines into sentiment, event type and impact score. Impact above 7 triggers a stress test.

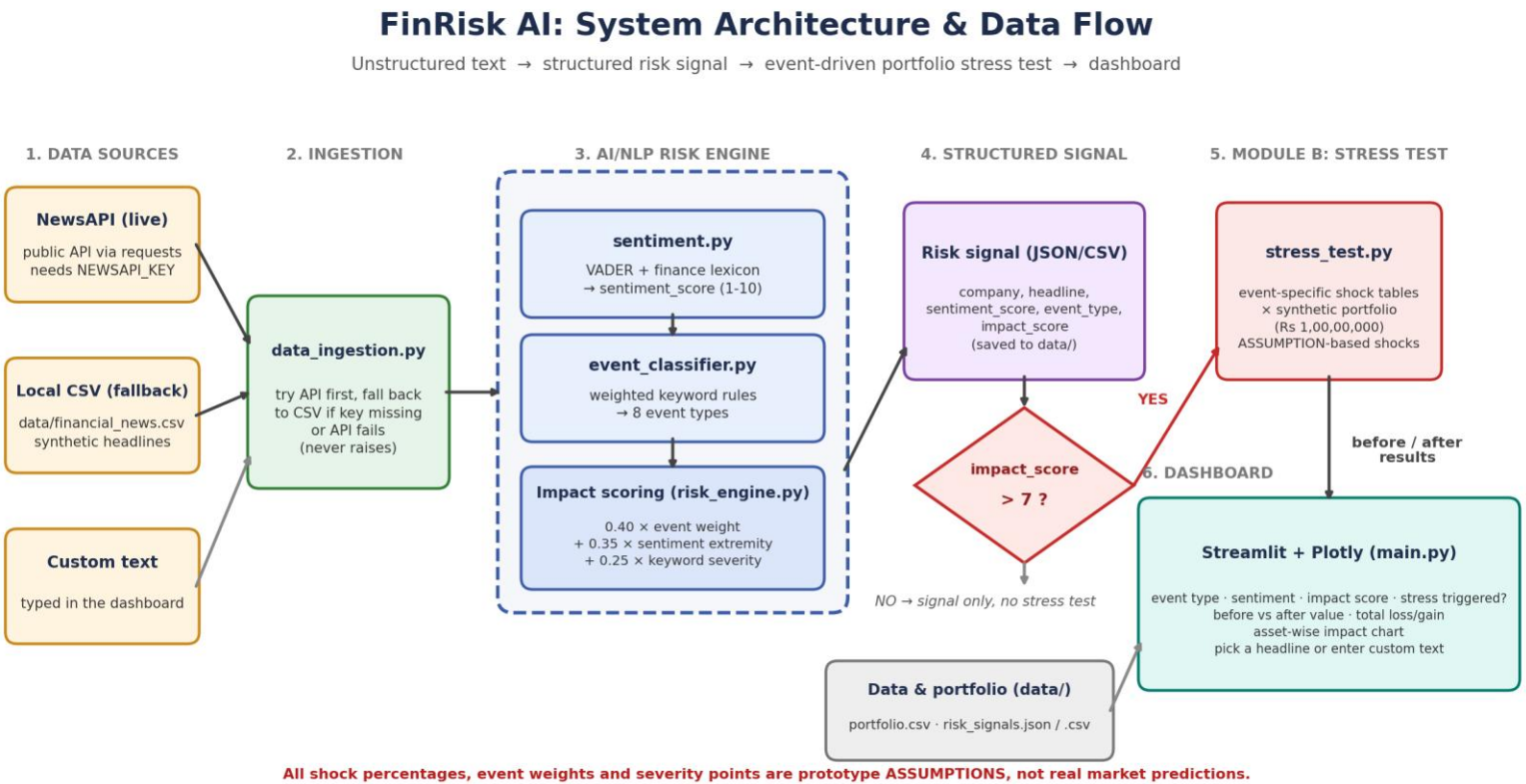
Scope

Unified NLP Risk Engine plus Module B, event-driven portfolio stress testing, built end to end.



Prototype ASSUMPTIONS: shock values, event weights and severity points are not real market predictions.

System Design



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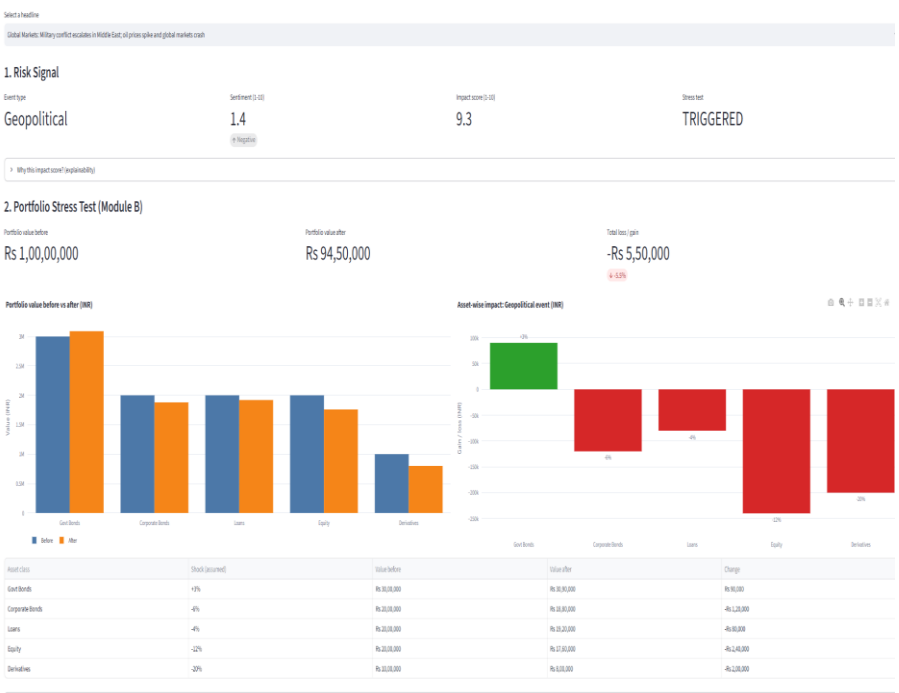
Implementation Highlights

Module	Tech choice	Why
Ingestion	NewsAPI + CSV fallback	Demo never fails offline
Sentiment	VADER + finance lexicon	Plain VADER misses "downgrade", "default"
Event & impact	Weighted keyword rules	Fully traceable, no black box
Stress test & UI	Shock tables, Streamlit, Plotly	Event-specific and interactive

impact = 0.40 × event weight + 0.35 × sentiment extremity + 0.25 × keyword severity

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Key Results



Headline	Event	Impact	Change
Military conflict, crash	Geopolitical	9.3	-Rs 5,50,000
Sanctions, supply fears	Geopolitical	8.4	-Rs 5,50,000
Yes Bank downgrade	Fin. Performance	7.7	-Rs 5,30,000
Recession, GDP falls	Macroeconomic	7.6	-Rs 5,20,000
SEBI probe, Adani	Regulatory	7.3	-Rs 4,50,000

Positive news (profit, launches) scored low and correctly did not trigger a stress test .20 headlines analyzed ->5 stress tests triggered.

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Domain Impact

1

Structured signals

Turns unstructured text into machine-readable risk signals other systems can consume.

2

Event to loss

Links an event to a portfolio-level loss figure for early warning and scenario talks.

3

Explainable by design

Every score shows its components, which suits audit and model-risk review.

4

Extensible

Ready for custom portfolios, live feeds and stronger models like FinBERT.

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Limitations & Next Steps

Limitations

- Rule-based NLP can miss sarcasm and negation
- No labelled-data validation, so no accuracy claim
- Shock tables and weights are assumptions
- One static synthetic portfolio

Next steps

- FinBERT or zero-shot classifier
- Validate on a Kaggle financial-news dataset
- Calibrate shocks with historical prices
- Custom portfolios and the GDELT feed

Thank you
DEMO VEDIO:https://youtu.be/6XBeXS6_SW4?si=slWE46omuLbArPBU
GITHUB LINK:https://github.com/hima-agarwal-11/FinRisk_AI

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